Homework 5: Message Passing Neural Networks

CS 1470/2470

Due November 13, 2019 at 11:59pm

1 Conceptual Questions

1. What is the benefit of using a Message Passing Neural Network over a vector representation of the graphs (e.g stacking all the features of the nodes/edges together indiscriminately) and a fully connected network? (3-4 sentences)

   Answer: Using a MPNN over a vector representation of the graphs totally ignores the structure of the data. A fully connected network could keep the local topological feature. Overall, using MPNN over a vector representation would end up with a lower accuracy rate than the fully connected network.

2. Describe how a message passing update is a generalization of a convolution. (2-3 sentences)

   Answer: Message passing on graph nodes is a generalization of convolution in several ways: 1. they both communicate with neighbors: convolution uses filter to communicate neighboring information of a pixel and massage passing on graph node uses a neural networks to pass node's feature information to neighboring nodes. 2. they both could use neighboring information to renew the node's feature: a filter could gather the value around the chosen pixel and renew the value, a message passing could receive the information sent by neighboring nodes and sum them all to renew. Sending and receiving information are called a round of propagation.

3. The purpose of the lifting function (known as f_init in the slides) is to transform the vertex/edge data into a new space. Why is operating in this space better than feeding raw features directly into the message passing network? (4-6 sentences)

   Answer: Just like embedding in recurrent neural network, we could use the lifting function which, essentially is a feed-forward neural networks to transform the raw features such as the atom index into a space that is more suitable express the node feature. Intuitively, the reason to lift

the raw data is to express physical or real data into the space where the neural network resides in. The real world data cannot be understood by the network unless doing such operation. And lifting function could also be helpful to improve the learning ability.

4. As mentioned in lecture, batching with an MPNN can be quite difficult. This is primarily due to the fact that you would need to trigger message passing across a number of graphs with varying numbers of nodes and edges. Still, even when we pad our graphs to all have the same number of nodes, a framework like DGL can only process a single graph at a time. Nevertheless, they still support batching. Give an explanation of how this could be done, and why it works (Hint: Think about the graphs in your batch as adjacency matrices). (3-5 sentences)

   Answer: Previously, mini-batching is a training method that splits training data into a number of small batches where each batch contains different training paired data. For graph neural network, the training data is a list of molecules where each molecule contains unstructured information of graph vertices and nodes values. But we could still split the list of molecules into small batches where each batch is simply a graph with a list of disconnected components. Each component is a molecule with connected vertices and nodes. Overall, this batch is still composed by disconnected componenets but this allows for the aggregation of parallelization of a bunch of small graph.

2 Ethical Implications

While this assignment focuses on molecular structure, it turns out that graphs are great representations for social networks, which makes social networks a very relevant application of deep learning on graphs (e.g. predicting influence or recommendations). (If you're curious, feel free to read about a variety of other cool applications here!)

Suppose you're working on such a system: given a social network represented as a graph, your task is to implement a neural net such that you can answer questions like: Who is likely to buy a product? Who are "influential" people in this network? What kind of messaging are parts of this network receptive to?

1. Why might an algorithm like this be useful? (1-2 sentences)

   Answer: Such neural net could help manufactures or retailer to target their customers more accurately. It could bring more interests to the company.

2. In what ways might this algorithm be dangerous? What kinds of questions might be asked–and by whom–that may be harmful, and what kind of harm could be caused? (2-3 sentences)

Answer: It can be really dangerous if people apply this network on some video or social networks. For example, if a running candidate wants to have more votes from his/her voters, he/she might be asking what preference do the voters want to see from him/her and how to affect their idea. This can be vividly showed by the episode "house of cards" where politician manipulate campaign process by back channeling social media.

3. Are there any ways to mitigate some of these harms, at least partially? (2-3 sentences)

   Answer: Strengthen online surveillance and legislation on internet oversee. Also the government or agency must be transparent enough to mitigate public concerns.

4. For you, personally, are there any boundaries you might draw in terms of what kinds of questions you're willing to ask (or answer)?

   Answer: Social network data could be really harmful and weaponized and it really depends on how people would like to use it. It is hard to say what kind of questions could be harmful. But personally, I will certainly not leak any customer-related information to anyone. And I will not ask any privacy related question to this network. I will only ask questions for research.